

properties. It is now used mostly as a blood thinner to avoid second (or first) heart attacks. The same thing applies to almost all drugs—many are used for secondary and tertiary properties.

I have just glanced at the desk in my business, nonliterary office (I separate the functional from the aesthetic). A laptop computer is propped up on a book, as I like to have some incline. The book is a French biography of the fiery Lou Andreas Salomé (Nietzsche's and Freud's friend) that I can very safely say I will never read; it was selected for its optimal thickness for the task. This makes me reflect on the foolishness of thinking that books are there to be read and could be replaced by electronic files. Think of the spate of functional redundancies provided by books. You cannot impress your neighbors with electronic files. You cannot prop up your ego with electronic files. Objects seem to have invisible but significant auxiliary functions that we are not aware of consciously, but that allow them to thrive—and on occasion, as with decorator books, the auxiliary function becomes the principal one.

So when you have a lot of functional redundancies, randomness helps on balance, but under one condition—that you can benefit from the randomness more than you can be hurt by it (an argument I call more technically *convexity to uncertainty*). This is certainly the case with many engineering applications, in which tools emerge from other tools.

Also, I am currently absorbed in the study of the history of medicine, which struggled under this Aristotelian illusion of purpose, with Galen's rationalistic methods that killed so many people while physicians thought they were curing them. Our psychology conspires: people like to go to a precise destination, rather than face some degree of uncertainty, even if beneficial. And research itself, the way it is designed and funded, seems to be teleological, aiming for precise results rather than looking for maximal exposures to forking avenues.

I have given more complicated names to this idea, in addition to *convexity*, like *optionality*—since you have the option of taking the freebie from randomness—but this is still work in progress for me. The progress coming from the second type of randomness is what I call *tinkering*, or *bricolage*, the subject of my next book.

Distinctions Without a Difference, Differences Without a Distinction

Another benefit of duplication. I have, throughout this book, focused on the absence of practical distinctions between the various notions of luck, uncertainty, randomness, incompleteness of information, and fortuitous